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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

STUDENT FEE PAYMENT TRACKING AND RECEIPT GENERATION SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled **“Student Fee Payment Tracking and Receipt Generation System”** submitted in partial fulfillment of the requirements for the course **B.Sc Information Technology** is a **record of my original work** carried out under the guidance of **Mr.R.Mugilan**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

DHARSHINI M

CERTIFICATE

This is to certify that the project entitled **"STUDENT FEE PAYMENT TRACKING AND RECEIPT GENERATION SYSTEM"** is a **bona fide work** carried out by **DHARSHINI M, I IT-B 2025**, **B.Sc Information Technology**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:	**Head of the Department**
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ABSTRACT

The **Student Fee Payment Tracking and Receipt Generation System** is a Java-based application designed to automate the process of managing student fee payments in educational institutions. In many institutions, fee collection and record maintenance are handled manually, which can lead to calculation errors, data inconsistency, delays, and difficulty in maintaining accurate records. This project aims to simplify and streamline the fee management process by providing an efficient and automated system.

The system allows users to enter student details, record fee payments, and automatically calculate the total fees paid along with any applicable charges or balances. It also generates a digital receipt for each transaction, ensuring accuracy and transparency. The application is developed using Core Java programming concepts, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented as a console-based application or with a graphical user interface using Java Swing.

The main objective of the system is to improve efficiency in fee management, reduce human errors, and maintain well-organized digital records of all transactions. The system generates a clear and structured receipt that includes student details, payment information, date, amount paid, and remaining balance if any.

This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a foundation for developing more advanced student management and financial systems in the future.

1. INTRODUCTION

In educational institutions, efficient fee management systems play a crucial role in maintaining accurate financial records and ensuring smooth administrative operations. Traditionally, many institutions rely on manual methods for collecting and recording student fees, where staff maintain handwritten records or use basic tools for calculations. This manual process is time-consuming and can often lead to calculation errors, delays, data loss, and difficulty in tracking payment history.

With the advancement of computer technology, automated fee management systems have become essential for improving efficiency and accuracy. A computerized system helps administrative staff easily record student payments, calculate totals automatically, and maintain well-organized digital records of all transactions.

The **Student Fee Payment Tracking and Receipt Generation System** is a Java-based application designed to automate the process of managing student fee payments and generating receipts. The system allows the user to enter student details, record fee payments, and automatically calculate the total amount paid along with any remaining balance. It also generates a clear and structured receipt that includes student information, payment details, date, amount paid, and balance if any.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a console-based program or using Java Swing for a graphical user interface (GUI).

The primary goal of this system is to reduce manual effort, improve accuracy in fee management, and provide faster and more reliable service in handling student transactions. This project also demonstrates how Java programming can be applied to develop simple real-world applications in the field of education and administration.

1.1 Objective

The main objective of the **Student Fee Payment Tracking and Receipt Generation System** is to develop a **simple and efficient application using Core Java** that **automates the process of managing student fee payments and generating receipts** in educational institutions.

The specific objectives of this project are:
• To **design and develop a Java-based student fee management system**.
• To **automate the calculation of total fees paid and remaining balance**.
• To **reduce manual errors in fee calculation and record maintenance**.
• To **provide a user-friendly system for staff to record payments and generate receipts quickly**.
• To **demonstrate the implementation of Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements.
• To **generate a clear and organized receipt for each student transaction**.

1.2 Overview

The Student Fee Payment Tracking and Receipt Generation System is a menu-driven application developed to help educational institutions manage student fee payments efficiently. The system provides a digital platform where administrators or staff can easily record, update, and manage student payment details in an organized manner.

The system allows the user to enter payment details such as student name, roll number, course, fee amount, payment date, and payment method. It stores all payment information in a structured format, making it easy to track records and manage financial data. The system also helps prevent errors such as duplicate or missing entries by maintaining proper records.

One of the important features of the system is the automatic receipt generation function, which generates a receipt immediately after the payment is recorded. This ensures transparency and provides proof of payment for students.

The system simplifies the fee management process by reducing manual record keeping and improving accuracy and efficiency. This project demonstrates the practical application of programming concepts in developing real-world software solutions for institutions that rely on systematic fee collection and tracking.

1.3 Features

The Student Fee Payment Tracking and Receipt Generation System includes the following main features:

• **Fee Payment Entry** – Allows staff to record student fee payments by entering details such as student name, roll number, course, amount, and payment method.

• **Payment Management** – Enables users to view, update, or delete payment records when required.

• **Receipt Generation** – Automatically generates a receipt after each successful payment for record and verification purposes.

• **Record Maintenance** – Stores all payment details in a structured format for easy access and future reference.

• **Search Functionality** – Allows users to search payment records using student name, roll number, or date.

• **User-Friendly Interface** – Provides a simple and easy-to-use interface for efficient fee management.

• **Error Reduction** – Minimizes manual errors by maintaining organized and accurate records.

• **Efficient Financial Tracking** – Helps institutions track fee payments and manage financial data effectively.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

In the Student Fee Payment Tracking and Receipt Generation System, the analysis involves studying how fee collection and record maintenance are currently handled in educational institutions. It helps in identifying the necessary features such as recording student details, tracking payments, generating receipts, and maintaining accurate financial records.

The analysis phase focuses on identifying the current issues in manual or semi-computerized fee management systems and proposing a computerized solution that improves efficiency, accuracy, and data management. This ensures better handling of student fee records, reduces errors, and simplifies the overall payment tracking process.

2.1 Problem Definition

In many educational institutions, student fee payments are often recorded manually using paper registers or simple spreadsheet methods. This traditional method creates several challenges.

Some common problems include:

- Errors in fee calculation or data entry
- Time-consuming fee recording and verification process
- Difficulty in managing payments of multiple students
- Lack of proper and organized financial records
- Delay in generating receipts for students
- Inefficient handling during peak fee collection periods

Manual fee management systems also increase the workload for administrative staff and can lead to confusion or dissatisfaction among students due to delays or incorrect payment records.

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing fee management system and proposing a computerized solution that improves efficiency, accuracy, and proper record maintenance.

2.2 Project Overview

The Student Fee Payment Tracking and Receipt Generation System is a Java-based application developed to simplify the process of managing student fee payments in educational institutions.

The system provides a menu-driven interface where users or administrative staff can enter student details and record fee payments easily. The system automatically calculates the total fee amount (if applicable) and generates a receipt after successful payment.

The system includes the following functionalities:

• Entering student details such as name, roll number, and course
• Recording fee payments and payment method
• Automatically calculating total fees or pending amounts
• Maintaining records of all transactions
• Generating receipts for each payment
• Viewing and managing student payment history

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application.**

2.3 Scope and Limitations

Scope

The scope of the Student Fee Payment Tracking and Receipt Generation System includes the following:

• Managing student fee details and records
• Recording and tracking fee payments
• Automatically calculating total fees and pending amounts
• Generating payment receipts for students
• Maintaining organized financial records
• Demonstrating the practical use of Java programming concepts

The system can be used in:

• Schools
• Colleges
• Universities
• Coaching centers
• Training institute

Limitations

• The system is limited to basic fee management and may not support advanced financial features

• It may not include online payment integration

• Data security features may be minimal in the basic version

• The system is designed for small to medium institutions and may not handle large-scale data efficiently

• Requires basic computer knowledge to operate

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Student Fee Payment Tracking and Receipt Generation System is designed as a menu-driven application where users can enter student details, record fee payments, and generate receipts automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Student Fee Payment Tracking and Receipt Generation System are explained below:

1. Student Details Module

This module is responsible for entering and displaying student information such as name, roll number, course, and contact details.

Functions of this module include:
• Accepting student details
• Storing student information
• Displaying stored records

2. Fee Entry Module

This module allows the user to record fee payments made by students.

Functions of this module include:
• Accepting payment details (amount, date, payment method)

• Recording fee transactions
• Storing payment information

The system allows entry of multiple student payments.

3. Fee Calculation Module

This module performs all fee-related calculations.

The system calculates:
• Total fee amount
• Paid amount
• Remaining balance (if any)

Example Calculation:
Total Fee = Course Fee
Balance = Total Fee – Paid Amount

4. Receipt Generation Module

This module generates a receipt after successful fee payment.

The receipt includes:
• Student name
• Roll number
• Course
• Amount paid
• Payment date
• Remaining balance

Example Receipt Output:

------ FEE RECEIPT ------

Student Name : Dharshini

Roll Number : 105

Course : B.Sc Information Technology

Amount Paid : 45000

Payment Date : 10/03/2026

Remaining Fee : 5000

Thank You!

5. Record Management Module

This module helps in maintaining and managing all student payment records.

Functions include:
• Viewing payment history
• Searching records
• Updating or deleting records

6. Exit Module

This module allows the user to exit the application safely after completing all operations.

When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Student Fee Payment Tracking and Receipt Generation System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:
• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1 Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of fee management in educational institutions.

The main problems identified in manual fee systems are:
• Manual calculations may lead to errors
• Recording payments manually takes more time
• Difficult to manage multiple student records
• Lack of proper tracking of fee payments
• Delay in generating receipts

Based on these problems, the system requirements were defined to develop an automated fee management system that simplifies payment tracking and receipt generation.

4.2 System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:
• Student Details Module
• Fee Entry Module
• Fee Calculation Module
• Receipt Generation Module
• Record Management Module
• Exit Module

The program flow and structure are designed so that users can easily interact with the system

4.3 Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:
• Creation of classes and methods
• Implementation of menu-driven logic
• Writing code for fee calculation
• Handling user input
• Generating and displaying receipts

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4 Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter student details	Details stored correctly
Enter fee amount	Amount accepted correctly
Calculate balance	Correct balance displayed
Generate receipt	Receipt generated correctly
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5 Result and Documentation

After successful testing, the results of the system are documented. The system generates receipts based on the fee payments entered.

The results are demonstrated through program execution and screenshots showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6 Algorithm

Algorithm for the Student Fee Payment Tracking and Receipt Generation System.

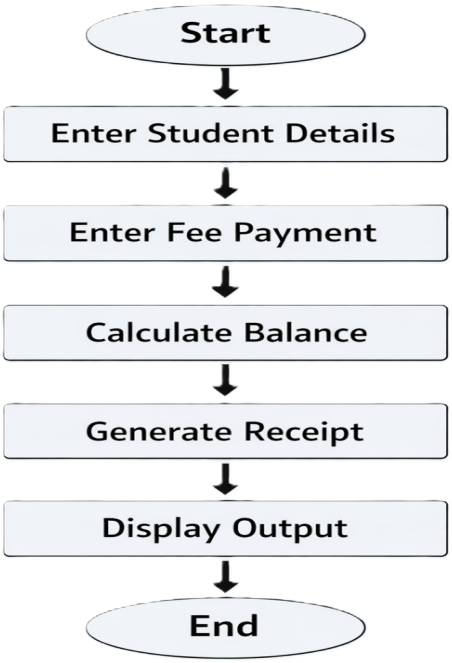
1. Start the program
2. Enter student details (name, roll number, course)
3. Enter fee amount and payment details
4. Calculate total fee and remaining balance
5. Store the payment record
6. Generate receipt

7. Display receipt to the user

8. Repeat for next student if required

9. End the program

4.7 Flowchart



5. SOURCE CODE

```
import java.util.ArrayList;
import java.util.Scanner;

class Student {
    String name;
    String rollNumber;
    String course;
    double totalFee;
    double paidFee;
    double balance;

    Student(String name, String rollNumber, String course, double totalFee, double paidFee) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.course = course;
        this.totalFee = totalFee;
        this.paidFee = paidFee;
        this.balance = totalFee - paidFee;
    }
}

public class FeeManagementSystem {

    static ArrayList<Student> students = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);
```

```
public static void main(String[] args) {
    int choice;
    do {
        System.out.println("\n===== Student Fee Management System =====");
        System.out.println("1. Add Student Fee Payment");
        System.out.println("2. View All Student Records");
        System.out.println("3. Search Student Record");
        System.out.println("4. Generate Receipt");
        System.out.println("5. Exit");
        System.out.print("Enter your choice: ");
        choice = sc.nextInt();
        sc.nextLine();

        switch (choice) {
            case 1:
                addStudent();
                break;
            case 2:
                viewAllStudents();
                break;
            case 3:
                searchStudent();
                break;
            case 4:
                generateReceipt();
                break;
```

```
            case 5:
                System.out.println("Exiting system. Thank you!");
                break;
            default:
                System.out.println("Invalid choice! Try again.");
        }
    } while (choice != 5);
}


static void addStudent() {
    System.out.print("Enter Student Name: ");
   String name = sc.nextLine();

    System.out.print("Enter Roll Number: ");
    String roll = sc.nextLine();

    System.out.print("Enter Course Name: ");
    String course = sc.nextLine();

    System.out.print("Enter Total Fee: ");
    double totalFee = sc.nextDouble();

    System.out.print("Enter Paid Fee: ");
    double paidFee = sc.nextDouble();
    sc.nextLine();
```

```
    double balance = totalFee - paidFee;
    System.out.println("Calculated Balance = Total Fee - Paid Fee = " + balance);

    Student student = new Student(name, roll, course, totalFee, paidFee);
    students.add(student);
    System.out.println("Student fee payment recorded successfully!");
}

static void viewAllStudents() {
    if (students.isEmpty()) {
        System.out.println("No records found.");
        return;
    }
    System.out.println("\nAll Student Records:");
    System.out.printf("%-15s %-10s %-20s %-10s %-10s %-10s\n",
            "Name", "Roll No", "Course", "Total Fee", "Paid", "Balance");
    for (Student s : students) {
        System.out.printf("%-15s %-10s %-20s %-10.2f %-10.2f %-10.2f\n",
                s.name, s.rollNumber, s.course, s.totalFee, s.paidFee, s.balance);
    }
}

static void searchStudent() {
    System.out.print("Enter Roll Number to search: ");
    String roll = sc.nextLine();
    boolean found = false;
    for (Student s : students) {
```

```
        if (s.rollNumber.equalsIgnoreCase(roll)) {
            System.out.println("Student Found!");
            System.out.printf("Name: %s\nRoll Number: %s\nCourse: %s\nTotal Fee:
%.2f\nPaid Fee: %.2f\nBalance: %.2f\n",
                    s.name, s.rollNumber, s.course, s.totalFee, s.paidFee, s.balance);
            found = true;
            break;
        }
    }
    if (!found) {
        System.out.println("Student record not found!");
    }
}

static void generateReceipt() {
    System.out.print("Enter Roll Number for receipt: ");
    String roll = sc.nextLine();
    boolean found = false;
    for (Student s : students) {
        if (s.rollNumber.equalsIgnoreCase(roll)) {
            System.out.println("\n------ FEE RECEIPT ------");
            System.out.println("Student Name   : " + s.name);
            System.out.println("Roll Number    : " + s.rollNumber);
            System.out.println("Course         : " + s.course);
            System.out.println("Total Fee      : " + s.totalFee);
            System.out.println("Amount Paid    : " + s.paidFee);

            double remainingFee = s.totalFee - s.paidFee;
```

```
            System.out.println("Remaining Fee  : " + remainingFee);


            System.out.println("-------------------------");
            System.out.println("Thank You!");
            found = true;
            break;
        }
    }
    if (!found) {
        System.out.println("Student record not found!");
    }
  }
}
```

6. INPUT AND OUTPUT SCREENS

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

D:\java class\project> javac  FeeManagementSystem.java

D:\java class\project> java  FeeManagementSystem

===== Student Fee Management System =====
1. Add Student Fee Payment
2. View All Student Records
3. Search Student Record
4. Generate Receipt
5. Exit
Enter your choice: 1
Enter Student Name: Dharshini
Enter Roll Number: 105
Enter Course Name: B.Sc-IT
Enter Total Fee: 50000
Enter Paid Fee: 45000
Calculated Balance = Total Fee - Paid Fee = 5000.0
Student fee payment recorded successfully!
```

```
C:\Windows\System32\cmd.e

===== Student Fee Management System =====
1. Add Student Fee Payment
2. View All Student Records
3. Search Student Record
4. Generate Receipt
5. Exit
Enter your choice: 2

All Student Records:
Name            Roll No    Course              Total Fee  Paid       Balance
Dharshini       105        B.Sc-IT             50000.00   45000.00   5000.00

===== Student Fee Management System =====
1. Add Student Fee Payment
2. View All Student Records
3. Search Student Record
4. Generate Receipt
5. Exit
Enter your choice: 3
Enter Roll Number to search: 105
Student Found!
Name: Dharshini
Roll Number: 105
Course: B.Sc-IT
Total Fee: 50000.00
Paid Fee: 45000.00
Balance: 5000.00
```

```
C:\Windows\System32\cmd.e

===== Student Fee Management System =====
1. Add Student Fee Payment
2. View All Student Records
3. Search Student Record
4. Generate Receipt
5. Exit
Enter your choice: 4
Enter Roll Number for receipt: 105

------ FEE RECEIPT ------
Student Name   : Dharshini
Roll Number    : 105
Course         : B.Sc-IT
Total Fee      : 50000.0
Amount Paid    : 45000.0
Remaining Fee  : 5000.0
-------------------------
Thank You!

===== Student Fee Management System =====
1. Add Student Fee Payment
2. View All Student Records
3. Search Student Record
4. Generate Receipt
5. Exit
Enter your choice: 5
Exiting system. Thank you!

D:\java class\project>
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The **Student Fee Payment Tracking and Receipt Generation System** was developed to automate the process of recording student fee payments and generating receipts. The system allows users to enter student details, record payments, calculate balances automatically, and generate fee receipts.

During the development process, **Core Java programming concepts** such as classes, objects, loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs fee calculations and generates accurate receipts.

The system was tested with different student inputs to ensure that it performs the required calculations correctly. The results show that the system reduces manual effort, improves record-keeping accuracy, and simplifies the management of student fee payments.

7.2 Conclusion

The **Student Fee Payment Tracking and Receipt Generation System** successfully demonstrates the practical application of **Core Java programming concepts** in developing a real-world business application for educational institutions.

The system automates fee tracking, reduces calculation errors, and provides faster service to both administrative staff and students. It simplifies student fee management by generating accurate receipts based on payment details and outstanding balances.

This project helps students understand how Java programming can be used to develop practical software systems. The system can be further enhanced by adding features such as **database connectivity, graphical user interfaces (GUI), online payment integration, and reporting tools** to make it more robust and user-friendly.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

1. https://www.oracle.com/java
2. https://www.geeksforgeeks.org
3. https://www.w3schools.com
4. https://www.javatpoint.com